Isaac Legene

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Education

Georgia Institute of Technology — Atlanta, GA
M.S. in Robotics - Electrical and Computer Engineering (GPA: 4.0) — *2025 – 2027*

Missouri University of Science and Technology — Rolla, MO
B.S. in Engineering Management - Industrial Engineering (GPA: 3.51) — *2022 – 2024*

Academic Experience

Georgia Institute of Technology: LIDAR Lab — Atlanta, GA
Graduate Researcher - Humanoid Whole-Body Loco-Manipulation and Tactile Sensing — *2025 – Present*

Diffusion Policy Fine-tuning for Humanoid Loco-Manipulation via Reinforcement Learning — *IEEE RA-L 2026*

- Trained and fine-tuned reinforcement learning and diffusion policies for humanoid loco-manipulation in Isaac Lab.
- Designed simulation environments and robot logic for data collection and policy training.

Tactile-Based Whole-Body Manipulation via Force-Supervised Contact-Aware Planning — *CoRL 2026*

- Participated in the design and implementation of tools for our in-house data collection interface.
- Coordinated, collected, and parsed human demonstration and teleoperated tactile data for policy training.

Georgia Institute of Technology: College of Computing — Atlanta, GA
Graduate Teaching Assistant – Machine Learning (CS 7641) — *Aug 2026 – Present*

Robotics and Automation Experience

Tesla — Austin, TX
Automation Controls Engineering Intern — *May 2026 – Aug 2026*
Material Flow Automation - Autonomous Mobile Robotics

- Commissioned and deployed autonomous mobile robots across multiple manufacturing production lines.
- Repaired and debugged AMR, PLC, and LiDAR scanner issues to support production operations.
- Developed scripts to auto-configure and push updates across the AMR fleet.

Rivian — Normal, IL
Automation Controls Engineering Intern — *Jan 2025 – Aug 2025*
Robotics simulation and virtual commissioning R2

- Developed full-scope production line simulations in Process Simulate.
- Supported design reviews for simulation and virtual commissioning of Rivian R2 production lines.
- Assisted the integration of a tool used to auto-generate collision zones in Process Simulate.
- Solely developed in a Unity simulation plugin realvirtual.io as a lower-cost alternative to Process Simulate.

Tesla — Fremont, CA
Automation Controls Engineering Intern — *Jan 2024 – Aug 2024*
Body-in-White Robotics & Production Equipment Integration for Models 3 & Y

- Oversaw and supported contract work and inventory for Body-in-White production equipment integration.
- Supported the integration of new production controls and processes to ramp up production.
- Implemented manufacturing electronics, HMIs, and PLC programs throughout multiple production lines.
- Completed training in Fanuc/KUKA (101–301), AC/DC circuits, multimeter usage, pneumatics, and PLC programming.

Dakkota Integrated Systems — Chicago, IL
Automation Controls Engineering Intern — *May 2023 – Aug 2023*
Tier 1 supplier for the Ford Expedition and Lincoln Navigator

- Assisted sustainability practices for all process automation systems to ensure quality and continuity of design.
- Troubleshoot and maintain network performance, connectivity, and hardware to mitigate future failures.
- Developed and updated standard work instructions for all manufacturing assembly lines and stations.

Roeslein & Associates — Saint Louis, MO
Electrical & Instrument Designer Intern — *May 2022 – Aug 2022*
Modular fabrication of biogas collection systems on farms domestically

Technical Skills

Robot Platforms: Kuka, Fanuc, AMRs, Unitree G1, T1 Booster, Dexmate
PLC and Programming Languages: C#/.NET, C++, Python, Allen-Bradley, Siemens
Software: Git, Process Simulate, Isaac Lab, ROS, Unity, Ignition, Linux, MuJoCo, Docker
Key Courses: Computer Vision, Computer Animation, Reinforcement Learning, Deep Learning, Machine Learning